

Build plan — 4 sessions, checkpoint + image evidence after each

Rule: every session ends with (a) a working, runnable checkpoint and (b) a screenshot saved under docs/img/. No session is "done" without the image.

#	Session	Builds	Image evidence
1	Real terrain field	Pull a real DEM of Lake San Antonio → derive sector graph (water/steep = gated ravine, edge cost = real slope) → swap the abstract grid in the working sandbox for the real field . Add the resource readout + 90-min phase clock.	live match running on the real hillshade, with sectors/bridges/squads
2	Game-design mechanics	Implement the design doc: classes (capture weights/abilities), intel/fog (see only scouted sectors), resource sinks (bridge/intel/siege), and the phase-gated 90-min arc (land-grab → contact → siege bell).	fogged map + class roster + phase banner + resource ledger
3	RL playtesters	Wrap the engine as a gym/PettingZoo	learning curve climbing + an RL

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		env with a microrts-style observation tensor + command action ; train PPO self-play at small scale (3v3, ~14 sectors); export trained-agent episodes.	episode replay + the strategy it found
4	Balance verdict + polish	OpenSpiel-style metrics (fairness, matchup matrix, exploitability), wire RL + heuristics into the Monte Carlo, link findings to the honesty ledger, polish UI, finalize the design doc.	the full "money-shot" screen with real data (not a mockup)

Each checkpoint is committable and shippable on its own — if we stop after any session, there's still a coherent artifact.

Known risk (Session 1): fetching real elevation data depends on network/tool access in the build environment. Primary source: USGS 3DEP via OpenTopography; fallback: AWS Terrain Tiles (Terrarium PNG → decode). If both are blocked, I'll use a real DEM file you drop in data/ — the pipeline is identical either way.